Informatics

Operational State Recognition and Braking Behaviour Analysis for Predictive Maintenance on Metro Trains

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Context & Problem

- Much rail maintenance runs on fixed intervals, regardless of how hard components were actually used. Predictive maintenance ties each intervention to measured condition instead.
- Metro friction brakes are a natural test case: the train brakes at almost every stop, hundreds of times a day, so wear should leave a large, repeating signal.
- **Data:** MetroAT, one year of 1 Hz recordings from a single Wiener Linien metro train (109 channels: speed, pneumatic system, mode signals), annotated with documented failure and maintenance events. All sensor values are min-max normalised, so the analysis works in relative units.

Research Questions

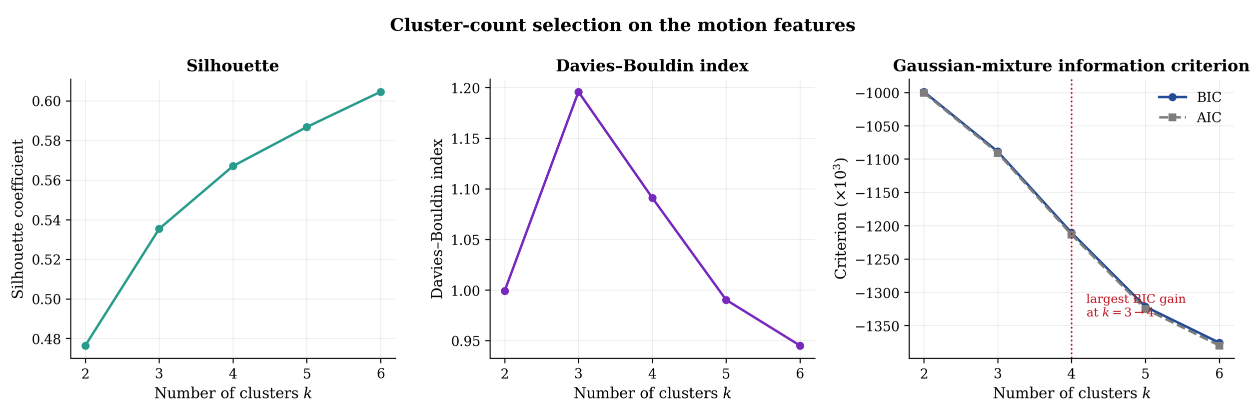
1. Do distinct operational regimes emerge from the motion data alone, without fixing their number in advance?
2. Can the air-system sensors recover the deceleration regime using only signals that did not help define it?
3. Is braking intensity a set of discrete classes or a continuum, and how much of it do the air-system sensors explain?
4. Does braking behaviour change detectably before documented brake-system failures?

Method

- 30 M raw rows ⟶ 1.49 M non-overlapping 10 s windows with per-channel summary statistics.
- Time-based split: training June 2024 to February 2025, held-out window February to June 2025. Every reported score is out-of-sample.
- **Leakage control by feature roles:** motion channels (speed, acceleration, jerk, deceleration energy) define the targets and are never predictors; only air-system channels (actuation + auxiliary) predict.
- 214 near-duplicate per-wagon sensor pairs (|r| > 0.98) absorbed by PCA instead of manual deletion.

Operational States

Motion-only k-means; the cluster count was selected by internal validation, not fixed in advance.

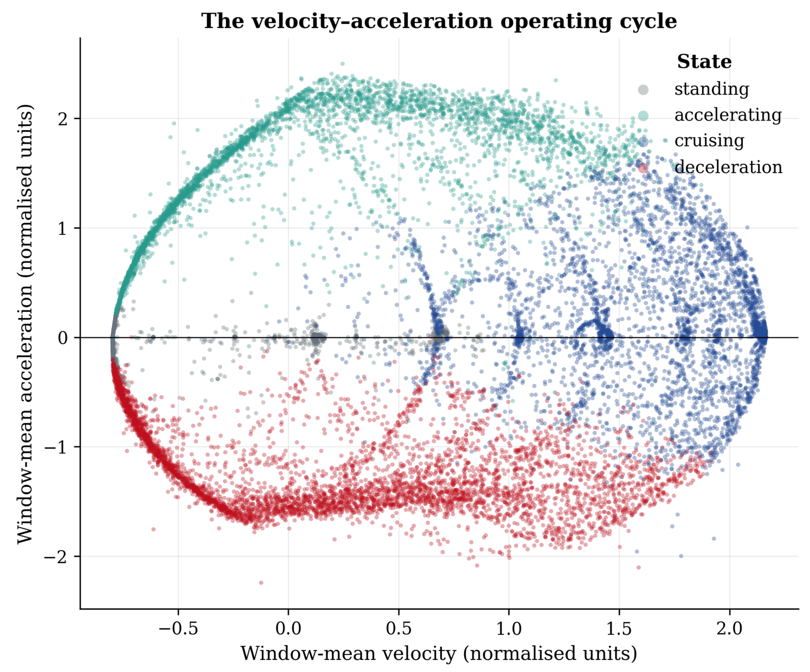


Cluster-count selection: the largest BIC improvement is the step k = 3 ⟶ 4; methods agree at k = 4 (ARI 0.77), stable under resampling (bootstrap ARI 0.87).

State	Share	Med. vel.	Med. acc.
standing	48.1 %	0.000	0.000
deceleration	20.3 %	0.343	−0.033
cruising	17.3 %	0.842	0.002
accelerating	14.3 %	0.305	0.041

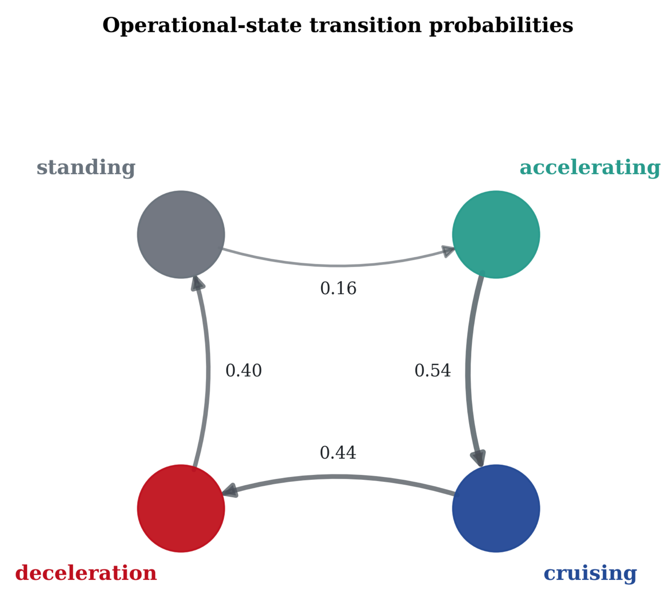
Plausibility of the Four States

- An earlier state definition that included brake pressures mislabelled a standing train holding its brake as "braking". Under the motion-only definition the deceleration state is 99.9 % genuine deceleration.
- A Random Forest given all features (0.994 accuracy) still ranks the kinematic ones first, which confirms the regimes are motion-defined rather than a pneumatic artefact.



The velocity-acceleration cycle, coloured by state: the train moves anticlockwise through stop, accelerate, cruise, decelerate. A minority of off-curve points motivate future work.

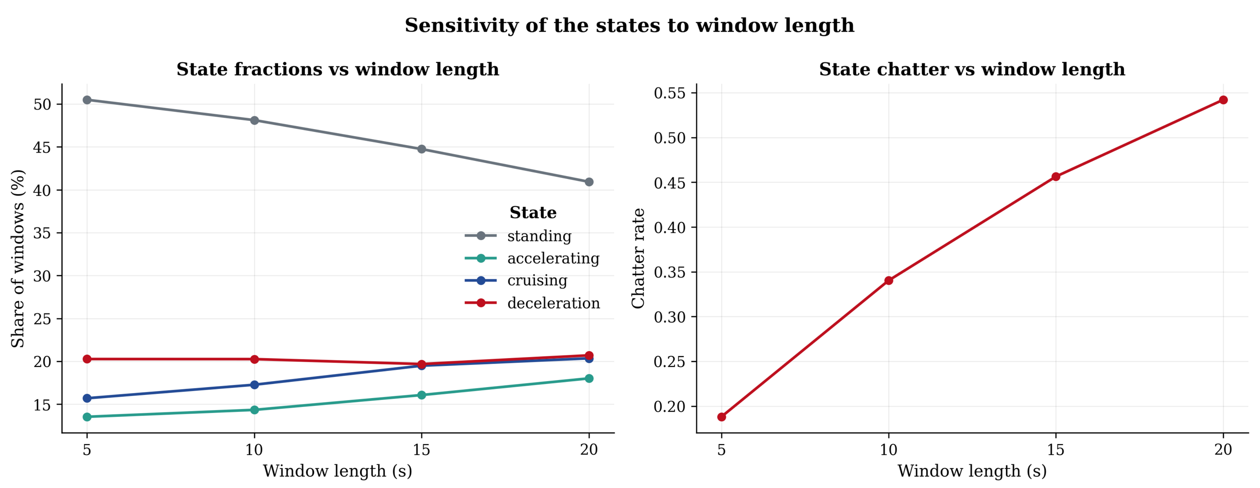
Transitions & Dwell



The transition matrix is dominated by the operating cycle stop ⟶ accelerate ⟶ cruise ⟶ decelerate ⟶ stop.

Mean dwell times match the physical picture: standing ≈ 54 s, accelerating ≈ 17 s, cruising ≈ 21 s, deceleration ≈ 24 s. The state mix is stable from month to month, so the regimes reflect how the train is operated rather than any particular period.

Window Sensitivity



Re-clustering at 5, 10, 15 and 20 s windows: state fractions are stable (deceleration ≈ 20 % throughout); label chatter rises with window length (0.19 ⟶ 0.54).

The 10 s window is a sound compromise between resolution and stability, and short windows are not the source of label churn.

Recovering Deceleration from Air Sensors

Can the pneumatic sensors, which never touched the state definition, recover the deceleration state on held-out data?

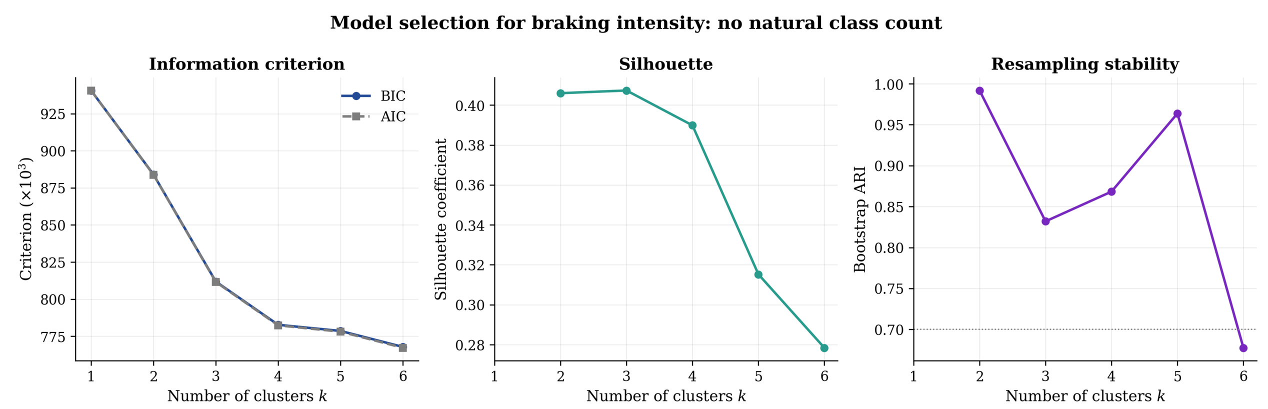
AUC 0.954	AUC 0.921
full tier random forest (all air-system channels)	auxiliary only (brake command withheld)

- Balanced accuracy 0.869, recall 0.956 for the deceleration class (full tier).
- Interpretable models agree on where the signal sits: pneumatic braking force and proportional-valve pressure dominate the decision tree and LDA.
- Deceleration leaves an independent fingerprint across the pneumatic system. The braking analysis rests on this premise.

Braking Events: Classes or Continuum?

A deceleration event is one continuous stretch of the deceleration state.

126,700	23.4 s	99.9 %
deceleration events in the training window	mean event duration (median 20 s)	genuine deceleration after removing held-brake windows

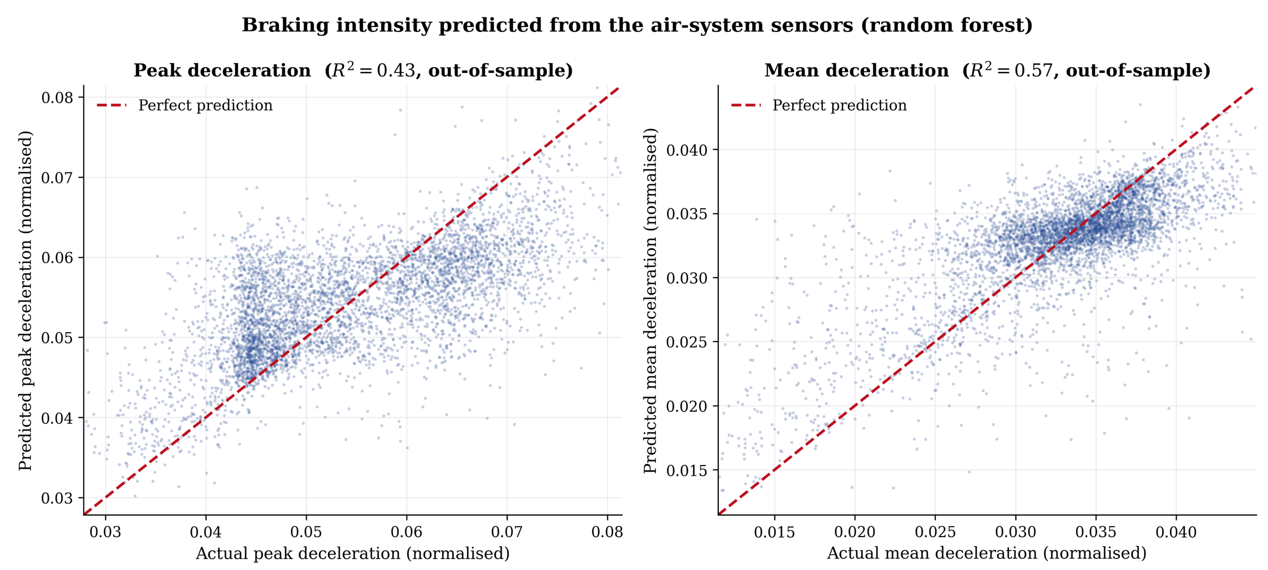


Model selection on intensity features: BIC keeps improving without settling, silhouette singles out no k, and the k = 5 stability bump only peels off a negligibly small outlier cluster.

- The intensity histogram shows a dominant mild mode and a broader firmer-braking hump, but they overlap without a clean gap and no clustering confirms the visual split.
- **Result:** braking intensity is a continuum, varying by degree rather than by kind. No light / hard / emergency labels are supported by the data; any such split is an operational threshold requiring domain input.

Intensity Regression & Feature Importance

Deceleration magnitude is predicted from the non-speed sensors on held-out events (baseline $R^2 \approx 0$).



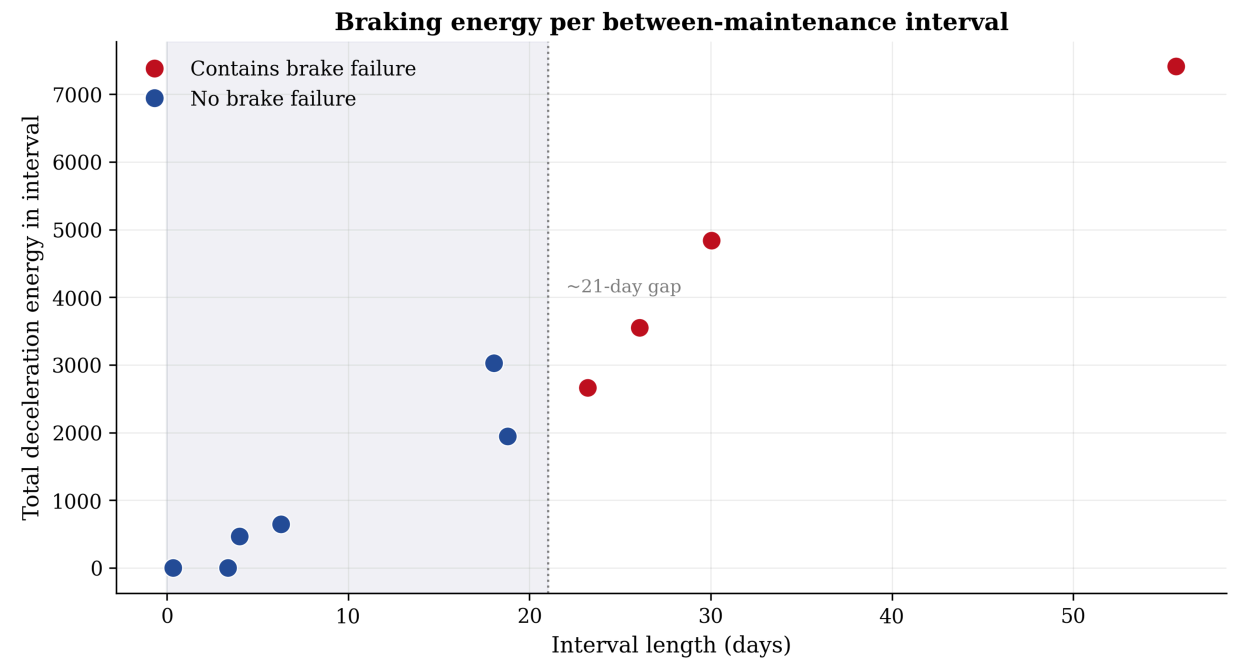
Predicted vs. actual intensity (random forest, full tier): positively associated, but compressed toward the mean. The pneumatic signal is real yet partial.

R^2 0.57	R^2 0.43	85 %
mean deceleration (RF); linear model: 0.20	peak deceleration (RF); linear model: 0.24	of specific braking energy ½(v_0^2 − v_1^2) explained

- A cross-validated Lasso keeps 19 of 22 air-system features across subsystems: the intensity signal is distributed, not carried by the brake command alone.

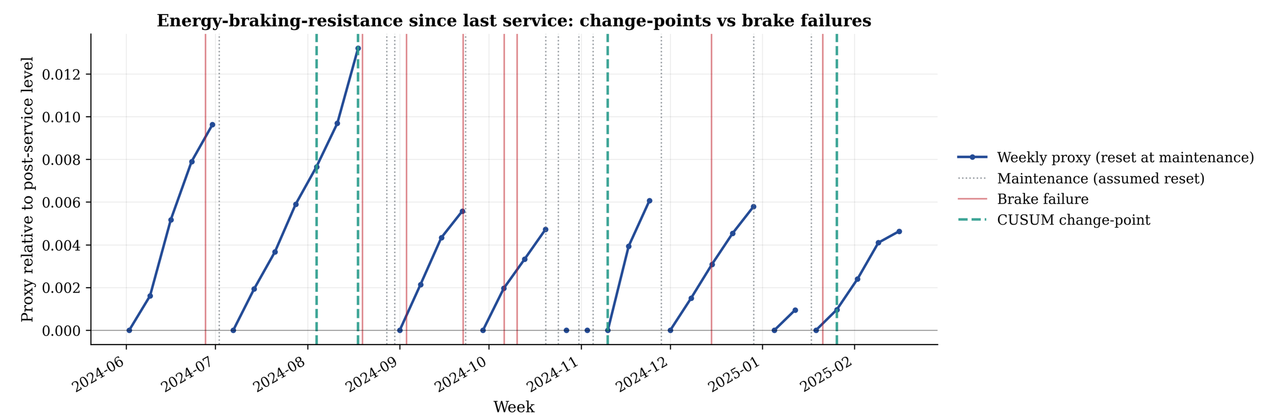
Maintenance & Failure Analysis

Braking energy vs. the maintenance calendar.



Deceleration energy summed per between-maintenance interval. ≈ 96 % of all real-deceleration energy falls inside the intervals. Every interval longer than ≈ 3 weeks contained a brake failure; every shorter one was failure-free. The split is suggestive, but it is confounded with exposure time and rests on ten intervals.

Change-point detection.



CUSUM on the weekly mean of energy-braking-resistance (strongest pre-failure channel), reset at each maintenance date: four change-points fire, only one within seven days of a documented failure.

Statistical null results.

- Pre-failure vs. baseline deceleration events (Mann-Whitney U, Bonferroni-corrected): p-values are significant only through sample size (> 23,000 events). Effect sizes are negligible: all braking-kinematic and brake-command features sit at Cliff's |δ| < 0.06, and only energy-braking-resistance reaches |δ| ≈ 0.20.
- Pre-event operational-state mix: 61 of 62 chi-square tests significant, but Cramér's V ≤ 0.07 (typically 0.018), and ≈ 0.003 once standing windows are excluded. The shift only reflects depot idling before scheduled work.
- With eight brake failures, no reliable pre-failure braking indicator exists in this data; the result is reported as a genuine negative.

Feature (pre-failure vs. baseline)	Cliff's δ	Above negligible?
energy-braking-resistance (mean)	−0.201	yes
load pressure (mean)	−0.103	no
jerk RMS	+0.054	no
brake-cylinder pressure (integral)	−0.040	no
peak deceleration	−0.005	no

Conclusions

- Motion alone yields four stable, physically meaningful operational states, robust to the window-length choice.
- The air system independently recovers deceleration (AUC 0.95; 0.92 without the brake command) and explains a moderate share of intensity and 85 % of dissipated energy.
- The link to failures is an honest null: no braking indicator separates the pre-failure week from baseline.
- **Next:** characterise the off-cycle outliers, compare wagons instead of collapsing them, and probe idle/charging behaviour, ideally on data with more documented failures.

Data: MetroAT (TU Wien Research Data, DOI 10.48436/9ja0q-bq581). Full reproducible pipeline: github.com/maxscheiblauer/Interdisciplinary_Project